C 42953 **(Pages : 4)** **Name.....**

Reg. No.....

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2023

Mathematics

MAT 2C 02—Mathematics

(2017—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type Questions)

Answer all *questions.*
Each question carries 1 *mark.*

1. Write $\sinh x$ in terms of exponential function.
2. If f is smooth on $[a, b]$, what is the length of the curve $y = f(x)$ from a to b.
3. Write the formula for n^{th} term of the sequence $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots\ldots$
4. Define geometric series.
5. Evaluate $\lim_{n \to \infty} (\ln n) \big/ n$.
6. What is the sum of the infinite series $1 + x + x^2 + x^3 + \ldots + x^n + \ldots., \mid x \mid < 1$.
7. Define power series about $x = a$.
8. Replace the Cartesian equation $x^2 + y^2 = 4$ with equivalent polar equation.
9. Find the Cartesian co-ordinate of the point $\left(\sqrt{2}, \frac{\pi}{4}\right)$ given in polar co-ordinates.
10. Find the domain of the function $f(x, y) = \sqrt{y - x - 2}$.

Turn over

11. Evaluate $\lim_{(x, y) \to (0, 0)} \frac{3x^2 - y^2 + 5}{x^2 + y^2 + 2}$.

12. Find $\partial f / \partial x$ and $\partial f / \partial y$, given $f(x, y) = x^2 - xy + y^2$.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 marks

13. Evaluate $\int_0^1 \sinh^2 x dx$.

14. Evaluate the improper integral $\int_0^1 \frac{dx}{\sqrt{1 - x^2}}$.

15. Find the limit of the sequence $\{a_n\}$, where $a_n = \frac{n + (-1)^n}{n}$.

16. Find the sum of the series $\sum_{n=1}^{\infty} \frac{7}{4^n}$.

17. Show that the series $\sum_{n=1}^{\infty} \frac{2n}{3n - 1}$ diverges.

18. For what values of x, the series $\sum_{n=0}^{\infty} x^n$ converges absolutely.

19. Replace the polar equations $r = \frac{4}{2 \cos \theta - \sin \theta}$ by equivalent Cartesian equations, and identify the graph.

20. Find a polar equation for the conic section with eccentricity $e = 1$ and with one focus at the origin and the directrix $x = 2$.

21. Find the area of the regions bounded by the circle $r = 2 \sin \theta$ for $\pi/4 \le \theta \le \pi/2$.

22. Find a spherical co-ordinate equation for the cone $z = \sqrt{x^2 + y^2}$.

23. Find $\partial^2 w / \partial x \partial y$ if $w = xy + \dfrac{e^y}{y^2 + 1}$.

24. Find dw/dt if $w = xy + z, \quad x = \cos t, \quad y = \sin t, \quad z = t$.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Derive the formula $\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right)$ for all real x.

26. Find the volume of the solid generated by revolving the region between the y-axis and the curve $x = 2/y, 1 \le y \le 4$, about the y-axis.

27. The line segment $x = 1 - y, 0 \le y \le 1,$ is revolved about the y-axis to generate the cone. Find its lateral surface area.

28. Investigate the convergence of the series $\sum_{n=1}^{\infty} \dfrac{(2n)!}{n!\,n!}$.

29. Find the sum of series $\sum_{n=1}^{\infty} \dfrac{4}{(4n-3)(4n+1)}$.

Turn over

30. For the series $\sum_{n=0}^{\infty} (-1)^n (4x+1)^n$.

(a) Find the radius and interval of convergence.

(b) For what values of x does the series converge absolutely ?

(c) For what values of x does the series converge conditionally ?

31. Find the Taylor series and the Taylor polynomials generated by $f(x) = e^x$ at $x = 0$.

32. Find the area inside the smaller loop of the limacon $r = 2\cos\theta + 1$.

33. Show that the function $f(x, y) = \frac{2x^2 y}{x^4 + y^2}$ is not continuous at the origin.

(6 × 5 = 30 marks)

Part D (Essay Type)

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. (a) Evaluate $\int_{-8}^{1} \frac{dx}{x^{1/3}}$. (5 marks)

(b) Find the volume of the solid generated by revolving about y-axis, the infinite region in the first quadrant between the curve $y = e^{-x}$ and the x-axis. (5 marks)

35. (a) Test for convergence the series $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdots\cdots (2n-1)}{4^n 2^n n!}$. (5 marks)

(b) Prove the series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n^2}\right)^n$ converges. (5 marks)

36. Find the linearization $L(x, y)$ of

$f(x, y) = x^2 - xy + \frac{1}{2} y^2 + 3.$

at the point $(3, 2)$. Find an upper bound for the error in the approximation $f(x, y) \approx L(x, y)$ over the rectangle $R: |x-3| \le 0.1, |y-2| \le 0.1$. Express the upper bound as a percentage of $f(3, 2)$, the value of f at the center of the rectangle. (10 marks)

[2 × 10 = 20 marks]